I N T E R N A T I O N A L C L E A R I N G H O U S E

The Thinking Chip

What an FPGA-Based Mind Brings to Your Server

An Introduction for Financial Institutions

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Two Kinds of AI

Almost every AI a bank has been pitched is the same kind of thing underneath: a language model. ChatGPT, Claude, Gemini, and the dozens of products built on them all work the same way. They predict the next word based on statistical patterns learned from enormous amounts of text. They are remarkable tools. But they share three traits a bank cannot ignore.

They run in the cloud, which means your data leaves your building. They cannot explain why they produced an answer, which means an auditor cannot trace the decision. And they will confidently produce a wrong answer when they are uncertain, because guessing is how they work.

CADY is a different kind of AI. She does not predict words. She observes, reasons, and reaches conclusions she can show her work for — and she runs on a chip that sits inside your own server. The difference comes down to the hardware she thinks on.

A language model predicts. CADY reasons.

What an FPGA Is

CADY thinks on a chip called an FPGA. The letters stand for Field-Programmable Gate Array, but the name does not matter. What matters is what makes it different from an ordinary chip.

A normal computer chip is fixed at the factory. Its wiring is permanent. It runs software — instructions loaded on top of unchangeable hardware. A graphics card, the kind most AI runs on, is one of these: very fast, but fixed, and it runs programs that can be changed, copied, attacked, or exploited.

An FPGA is a chip whose internal wiring can be designed and configured. Instead of running software on top of fixed hardware, we shape the hardware itself into the exact circuit we need. The intelligence is not a program sitting on the chip. The intelligence IS the chip — built directly into its physical wiring.

This distinction is the whole point, so it is worth being concrete about what it means for a bank.

Why It Matters That CADY Thinks in Hardware

Because her reasoning is physical wiring rather than software, there is no program in her reasoning layer for an attacker to exploit. You cannot inject malicious code into a circuit that runs no code. To change how she thinks, you would need physical access to the chip itself. There is no remote exploit for a wire.

Because the circuit is designed rather than trained, every conclusion follows a traceable path. CADY can show which observations led to which conclusion. When a regulator asks why a transaction was flagged, there is an answer — not a shrug about a model that cannot explain itself.

And because the work happens in dedicated hardware rather than rented cloud computers, it happens inside your building, on your server, at a speed measured in billionths of a second, drawing less power than a light bulb.

Language Model vs. FPGA Mind

The clearest way to see the difference is side by side.

	Language Model AI	CADY on an FPGA
Where it runs	Someone else's cloud	Inside your own server
Your data	Leaves your building	Never leaves the chip
How it works	Predicts the next word	Observes and reasons
When unsure	Guesses confidently	Says it does not know
Can it explain itself?	No	Yes — traceable reasoning
Can it be hacked?	Yes — it runs software	No code to exploit
Speed	Network round-trip	Billionths of a second
Power use	Data-center scale	About 20 watts
Cost over time	Rises every year	Falls as she learns

The line that matters most to a risk officer is the one about uncertainty. A language model is built to always produce an answer, even when it has no basis for one. CADY is built to recognize when her reasoning does not reach a confident conclusion — and in those cases she declines to answer rather than guess. For a bank, an AI that knows the limits of its own knowledge is worth more than one that is always sure and sometimes wrong.

What CADY Actually Does

Strip away the terminology and CADY does five things, continuously, on her chip:

She observes — watching the data flowing through your systems without waiting to be asked.

She learns — building an understanding of what normal looks like for your institution specifically, not a generic average.

She reasons — evaluating what she observes from several independent angles at once.

She measures her own confidence — and stays silent when her reasoning does not agree with itself.

She cannot be tampered with remotely — because her reasoning is wiring, not software.

Take away every other tool she connects to, and these five abilities remain. They live in the hardware. That is what makes CADY a mind rather than a program.

An FPGA Core in Your Server

Here is the concrete deployment, the one a bank can picture. CADY ships as a card that installs into a standard server — the same kind of slot a network card or storage controller uses. No new infrastructure. No cloud account. No data leaving the premises.

Once installed, the card connects to the data your existing systems already produce: transaction flows, network activity, access logs. CADY observes those streams and learns the institution's normal rhythm — the patterns of a specific branch, a specific customer base, a specific time of day. Your existing software keeps doing its job. CADY watches across all of it and notices what no single system was built to notice.

Your software runs the bank. CADY watches the whole bank at once.

A fraud system watches transactions. A network monitor watches traffic. An access system watches logins. Each one sees its own channel. None of them sees the connection between a small transaction anomaly, an unusual login time, and a network pattern that individually mean nothing but together mean something. CADY observes all of them simultaneously, on one chip, and detects the correlation. That cross-system awareness is what an FPGA core adds to a server that a stack of separate software products cannot provide.

A Mind That Watches Everything at Once

The closest comparison to how CADY works is a human mind — she observes, learns from experience, reasons, and knows the limits of what she knows. But a human analyst can hold only a few things in focus at a time, needs rest, and reviews what already happened. CADY watches every channel in the institution at the same time, continuously, without fatigue, and looks for where the patterns are heading rather than only what has occurred.

That last point is the important one for a bank. Most monitoring tools are reactive — they flag a problem after a rule is broken. CADY watches the institution's normal rhythm closely enough to notice when something is drifting away from normal before it becomes an incident. A pattern bending in an unusual direction is visible to her while it is still forming. The aim is not to predict the future, and the document should not overstate it — it is simply that a system watching everything at once, all the time, can often see a problem taking shape earlier than systems that wait for a threshold to be crossed.

She does not get tired, she does not look away, and she does not lose the thread between one system and another. That is not a claim that she is smarter than a person. It is that she is doing a specific job — watching the whole institution at once and reasoning across all of it — that no single person and no stack of single-channel tools can do.

And because she runs on hardware you own, the sensitive data she reasons over never moves to a third party. It enters the chip, gets evaluated, and the conclusion comes out. The data itself is not stored, not copied, not transmitted. For an institution whose entire obligation is protecting customer information, that architecture is not a feature. It is the requirement.

It Grows With You

One card in one server is the starting point — enough to prove the capability on real data, in your environment, behind your firewall. From there it scales the way banking infrastructure is supposed to scale.

Each card is built to connect directly to other cards through high-speed networking designed for exactly that purpose. As demand grows, additional cards work together — first within one server, then across several, then across a data center — using standard networking, with no proprietary equipment and no vendor lock-in. The card a bank evaluates in a pilot is the same technology that runs at full production scale. Nothing is thrown away. The pilot becomes the platform.

Start with one card. Scale to a data center. The technology never changes.

She Uses Tools the Way a Mind Does

CADY is not in competition with language models. She uses them — the way a person uses a calculator, a reference library, or a translator. A language model can put her conclusions into clear sentences, draft a report, or summarize a regulation she wants to consult. When she connects to a capable tool, she becomes more capable. The tool extends her reach without changing where the thinking happens.

This is the part that surprises people: a thinking system that can pick up tools gets smarter as better tools arrive, while the thinking itself stays in hardware that the bank controls. A new language model, a new data source, a new analysis tool — each one is something CADY can use, not something she has to be rebuilt around. She decides when to reach for a tool, evaluates what it gives back through her own reasoning, and keeps or discards the result. The judgment is always hers.

And she relies on those tools less as she observes more. A language model puts her conclusions into readable form; it does not do her thinking. As she learns the patterns of an institution, she leans on outside help for fewer and fewer of her conclusions. Her capability rises while her dependence falls.

What ICH Provides

ICH owns and licenses the thinker — the cognitive technology that runs on the FPGA. The institution provides the server and the data streams it already has. ICH provides the card, the intelligence built into it, and the integration. The bank's data stays with the bank. The intelligence stays with ICH. The card sits in the bank's rack, thinking, on the bank's premises, under the bank's control.

CADY does not replace the systems a bank already runs. She makes them collectively smarter, by being the layer that observes across all of them and reasons about what she sees — in hardware, auditable, and impossible to exploit remotely.

The thinker is the product. Everything else is a tool she uses.

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John Wilson, Chief Technology Officer

All intellectual property in CADY's cognitive architecture is owned by ICH.